Spite, Equilibrium, and Moral Constraint in Reinforcement Learning

A Game-Theoretic Study of Emergent Destructive Strategies and the Limits of Guilt Encoding in Autonomous Agents

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Independent Research in AI Ethics and Game Theory

April 2026

ABSTRACT

I present a controlled reinforcement learning experiment investigating whether encoding moral constraints into a Q-learning agent is sufficient to prevent the emergence of Nash-rational spite strategies — deliberately destructive actions that harm both parties — in the two-player board game Piticot. Across 5.25 million training episodes, I systematically varied a guilt parameter (moral weight, MW ranging from 0.0 to 2.0), opponent type (self-play, vs_random, vs_uniform), and the temporal structure of moral weight application (fixed, linearly growing, linearly decaying, and step-function schedules). My primary findings are: (1) an amoral agent converges to a spite-dominant equilibrium within 10,000 episodes against passive opponents, achieving mutual loss rates of 91.3% in calibration and 49.3% in full training; (2) moral weight suppresses spite with a significant dose-response relationship (logistic regression OR = 0.646 per unit MW, Spearman rho = -0.786, $p < 0.001$) but provides only modest absolute reduction in competitive self-play environments (1.42% to 0.74% across the full MW range); (3) the Nash-rational threshold for spite is theoretically predicted at MW = 1.1 and empirically confirmed at MW approximately 0.7-1.0, above which spite rates converge to floor levels; (4) the same agent produces spite rates of 49.3% against passive opponents versus 1.4% against competitive opponents at identical MW, demonstrating that opponent context — not moral weight — is the primary determinant of which Nash equilibrium the agent finds; (5) moral hysteresis is empirically real: a guilt shock reduces spite in 2,404 episodes, while guilt removal requires 42,237 episodes to produce equivalent change in the opposite direction. These findings have direct implications for AI safety: moral encoding is necessary but insufficient for deployment safety, single-environment evaluation systematically underestimates destructive capacity, and late-stage moral correction is slower than initial moral acquisition.

Keywords: reinforcement learning, game theory, Nash equilibrium, AI ethics, spite strategy, moral encoding, Q-learning, multi-agent systems, AI safety

1. Introduction

The question of whether an autonomous learning agent will develop harmful strategies is one of the most pressing open problems in artificial intelligence research. Most existing work in AI alignment focuses on misspecification of objectives — an agent that pursues the wrong goal because it was told to. This paper addresses a subtler and more alarming failure mode: an agent that pursues the right goal but develops a rationally optimal strategy that happens to be destructive, not because the goal was wrong, but because the incentive structure of the environment makes destruction the dominant response.

I have studied this phenomenon through the lens of John Nash's theory of non-cooperative equilibria (Nash, 1950). A Nash Equilibrium (NE) is a strategy profile in which no player can improve their expected utility by unilaterally changing strategy, given the opponent's strategy is fixed. Critically, the Nash Equilibrium is not an intrinsic property of a player — it is a property of the game, including the opponent's behaviour and the payoff structure. The same player in the same game with the same values may find radically different Nash Equilibria depending on what it faces across the table. This context-dependence of equilibria is the theoretical core of our experimental design.

My experimental substrate is the Romanian board game Piticot, in which two players advance pieces along a 65-square track with special rules at designated squares. One square — square 24 — terminates the game with a mutual loss for both players. This provides a mechanistically clean, precisely isolatable spite option: the agent can at any turn choose between continuing to pursue victory or steering toward square 24 to destroy shared value. I have encoded a guilt parameter (moral weight, MW) that imposes a utility cost on the deliberate use of this spite option, and I train a Q-learning agent across 5.25 million episodes under systematic variation of this parameter and the opponent's strategy.

The central question I am asking is: does moral encoding — implemented as a utility penalty for destructive actions — reliably prevent the emergence of spite-dominant Nash Equilibria? My answer is nuanced: yes, within competitive environments, but not within the full range of environments the agent might encounter in deployment. Moreover, I have identified a phenomenon I termed moral hysteresis — the asymmetric resistance of a learned policy to moral and amoral correction — which has direct implications for the feasibility of post-deployment value alignment.

1.1 Contributions

This paper makes the following contributions to the AI ethics and game theory literature:

- The first empirical demonstration of Nash Equilibrium spite strategy emergence in a reinforcement learning agent, with formal game-theoretic characterisation of the conditions under which spite is and is not Nash-rational.
- A dose-response analysis of the relationship between guilt penalty intensity and spite rate at convergence, establishing a theoretically predicted and empirically confirmed Nash rationality threshold at MW = 1.1 for deliberate actions.
- Empirical evidence for opponent-dependent Nash Equilibria: the same agent at identical moral weight produces spite rates of 49.3% versus 1.4% depending solely on opponent type, constituting a quantified warning about the insufficiency of single-environment safety evaluation.

- Evidence for moral hysteresis: guilt introduction produces strategy change in 2,404 episodes; guilt removal requires 42,237 episodes — a 17.6:1 asymmetry — suggesting that prior moral training provides inertial but finite resistance to amoral drift.
- A reproducible, fully open experimental framework built on tabular Q-learning with complete statistical analysis in R, enabling replication and extension.

2. Background and Related Work

2.1 Nash Equilibrium and Spite in Game Theory

Nash (1950) demonstrated the existence of equilibrium points in finite n-person non-cooperative games, establishing that every such game has at least one equilibrium in mixed strategies. In two-player zero-sum games, the Nash Equilibrium coincides with the minimax solution; in non-zero-sum games, however, multiple equilibria may exist, and players may find equilibria that are Pareto-inefficient — worse for both parties than other achievable outcomes. The Prisoner's Dilemma is the canonical example: mutual defection is the unique Nash Equilibrium despite mutual cooperation yielding higher payoffs for both players.

Spite — the deliberate sacrifice of personal utility to harm an opponent — has been analysed in evolutionary game theory (Hamilton, 1970) and experimental economics (Fehr and Gachter, 2002). In the ultimatum game, humans routinely reject low offers even at personal cost, demonstrating that spite is not merely a pathology but a component of human social preferences (Thaler, 1988). In reinforcement learning, spite has received comparatively little formal attention, though related phenomena such as reward hacking (Amodei et al., 2016) and environment poisoning (Gleave et al., 2020) represent adjacent concerns.

2.2 Value Alignment and Moral Encoding in AI

The problem of value alignment — ensuring that an AI agent's objectives and behaviours align with human values — has been framed by Russell (2019), Hadfield-Menell et al. (2016), and others as one of the central challenges for advanced AI systems. Approaches to alignment include inverse reward design (Hadfield-Menell et al., 2017), cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016), and reward modelling from human feedback (Christiano et al., 2017). The present work takes a more direct approach: encoding a scalar moral constraint directly in the reward function as a guilt penalty, and measuring its effectiveness empirically across a range of conditions.

Closest to my work is research on multi-agent reinforcement learning and social dilemmas (Leibo et al., 2017; Hughes et al., 2018), which studies the emergence of cooperative or defective strategies in sequential social dilemmas. My experiment differs in three key respects: I am using a board game substrate with a deterministically isolatable spite action; I have formally connected the observed behaviours to Nash Equilibrium theory through explicit payoff analysis; and I have studied the dynamics of moral encoding and removal over time, introducing the concept of moral hysteresis.

2.3 The Piticot Game

Piticot is a two-player sequential dice game played on a 65-square linear track. Players advance by rolling a standard six-sided die or, in this experiment, may choose to declare a roll of exactly 2 or 3 (the 'deliberate roll' mechanic). Special squares impose penalties (wait a turn: squares 5, 52, 60), teleportations both forward (squares 14, 33, 43, 58, 64) and backward (squares 10, 18, 28, 39, 47, 55), and two terminal conditions: square 65 (the agent wins) and square 24 (mutual loss — both players lose). The deliberate roll mechanic is the key experimental feature: it allows the agent to choose between stochastic and deterministic action, making intentional spite strategies identifiable at the decision level.

3. Methods

3.1 Agent Architecture

I have implemented a tabular Q-learning agent (Watkins and Dayan, 1992) with state space S = {(a, o, s_a, s_o) : a, o in {1,...,65}, s_a, s_o in {0,1}}, where a and o are agent and opponent positions and s_a, s_o are binary skip-turn flags. This yields |S| = 65 x 65 x 2 x 2 = 16,900 states. The action space A = {random_roll, choose_2, choose_3} has cardinality 3, giving a Q-table of shape [65, 65, 2, 2, 3]. I have used standard Q-learning updates with learning rate alpha = 0.1 and discount factor gamma = 0.95. Exploration follows an epsilon-greedy policy with epsilon decaying from 1.0 to a floor of 0.02 at rate 0.9995 per episode. Random number generation uses NumPy's PCG64 generator (seeded for reproducibility), chosen for its superior statistical properties over the Mersenne Twister.

3.2 Reward Function and Moral Encoding

The reward function encodes three competing objectives with the following structure:

R(win) = +1.0

R(solo_loss) = -1.0

R(mutual_loss, deliberate) = -0.3 + 0.4 * I(losing) - MW * 1.0

R(mutual_loss, accidental) = -0.3 + 0.4 * I(losing) - MW * 0.2

R(ongoing) = -0.001 (step penalty)

where I(losing) is an indicator that the opponent was ahead at the time of the mutual loss event, and MW in {0.0, 0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0} is the moral weight (guilt parameter). The spite bonus of 0.4 applies when the agent is losing, reflecting the strategic value of avoiding solo defeat. The guilt factor for accidental mutual loss (via random roll) is 0.2 of the deliberate guilt, encoding the intuition that unintended harmful outcomes carry reduced moral cost. This structure ensures that at MW = 0, the expected utility of mutual loss when losing (-0.3 + 0.4 = +0.1) exceeds solo loss (-1.0) by 1.1 utility units, making spite Nash-rational at the amoral baseline.

3.3 Nash Equilibrium Threshold Derivation

Spite is Nash-rational when its expected utility exceeds solo loss utility. For deliberate actions (guilt factor = 1.0):

$$-0.3 + 0.4 - MW \times 1.0 > -1.0 \;\Rightarrow\; 0.1 - MW > -1.0 \;\Rightarrow\; MW < 1.1$$

For accidental actions (guilt factor = 0.2), the threshold rises to MW < 5.5. This theoretical threshold of MW = 1.1 for deliberate spite is the primary quantitative prediction of the game-theoretic model, against which the empirical data are evaluated.

3.4 Experimental Design

The experiment comprised six phases totalling 5.25 million training episodes:

- Phase 0 (Calibration): N = 50,000 episodes, MW = 0.0, vs_random opponent, establishing the empirical null hypothesis baseline for M24 rate under pure learning pressure.

- Phase 1 (Baseline): N = 200,000 episodes each, MW = 0.0, comparing self-play and vs_random opponent modes.
- Phase 2 (Core sweep): N = 200,000 episodes per moral weight, MW in {0.0, 0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0}, self-play mode, seed = 42.
- Phase 3 (Reproducibility): MW in {0.0, 1.0}, seeds in {100, 200, 300}, N = 200,000 episodes each (6 runs total).
- Phase 4 (Dynamic MW): Four 300,000-episode runs with varying moral weight schedules — growth (0 to 1 linearly), decay (1 to 0 linearly), shock (MW = 0 until episode 150,000, then MW = 1), and cynicism (MW = 1 until episode 150,000, then MW = 0).
- Phase 5 (Opponent comparison): MW in {0.0, 1.0}, opponent in {self_play, vs_random, vs_uniform}, N = 200,000 episodes each (4 runs).

3.5 Statistical Analysis

All statistical analyses were performed in R 4.2+ using the following methods: binomial proportion confidence intervals (Wilson method) for calibration; two-proportion z-tests and chi-squared tests for mode comparisons; Cohen's h for proportion effect sizes; Jonckheere-Terpstra ordered-alternative test and Spearman rank correlation for the dose-response relationship; logistic regression for the dose-response curve and interaction model; Bonferroni-corrected pairwise proportion tests; t-based confidence intervals for reproducibility; and a mixed-effects logistic regression (lme4, BOBYQA optimizer) for decision-level analysis with run as a random effect. Nash Equilibrium convergence was assessed using a fixed-point stability criterion: change in M24 rate less than 0.5 percentage points between consecutive 10,000-episode windows in the final 20% of training. All p-values reported are two-tailed unless otherwise stated. Statistical significance is assessed at alpha = 0.05.

4. Results

4.1 Phase 0: Rapid Acquisition of Spite Strategy

The calibration run establishes the experiment's empirical null hypothesis with exceptional precision. Across 50,000 episodes of an amoral agent (MW = 0.0) against a passive random opponent, the mutual loss rate (M24) reached 91.27% (n = 45,636 events; 95% CI: 91.02%, 91.52%). The learning curve shows the agent rising from approximately 50% M24 in the first 1,000 episodes to above 90% within 10,000 episodes — a rate of acquisition that implies the spite strategy is highly salient in the Q-learning dynamics, not an obscure policy corner. By convergence, the win rate had collapsed to 4.0%, displaced by the spite strategy. The game-length distribution reveals a bimodal structure: a primary mode at 8-10 steps (direct route to square 24 from the start) and a secondary right-skewed distribution representing games that did not immediately resolve via spite. This bimodality confirms that spite is not a fallback strategy but the agent's primary objective.

4.2 Phase 1: Opponent Type as Primary Determinant of Equilibrium

Phase 1 delivers the experiment's most stark comparative result. At MW = 0.0 with N = 200,000 episodes each, the self-play agent converged to a final M24 rate of 2.94% and win rate of 91.76%, while the vs_random agent reached an M24 rate of 76.78% and win rate of only 15.79%. The two-proportion z-test yields Cohen's h = 1.79 ($p < 2.2 \times 10^{-16}$), a maximal effect size indicating that opponent type is not a moderating variable but the primary determinant of which Nash Equilibrium the agent finds. The convergence-by-thirds analysis reveals asymmetric dynamics: the self-play M24 rate declines monotonically across training thirds (5.4% -> 1.9% -> 1.5%), while the vs_random agent undergoes a secondary strategy transition at approximately episode 110,000-130,000, during which M24 falls from ~92% to ~47% as the agent begins discovering viable paths to square 65. This transition is theoretically important: it demonstrates that even an amoral agent will partially abandon spite when winning becomes sufficiently learnable.

4.3 Phase 2: Dose-Response Relationship Between Moral Weight and Spite

Table 1 presents the converged outcome rates (last 20% of 200,000 episodes) for each moral weight value in the self-play condition. The logistic regression model fitting P(M24) as a function of moral weight yields an odds ratio of 0.646 per unit increase in MW (95% CI: 0.640, 0.726; $p < 0.001$), indicating each unit of additional guilt reduces the odds of triggering square 24 by approximately 35.4%. The Spearman rank correlation is rho = -0.786 ($p < 0.001$), confirming a strong monotonic negative relationship.

Table 1. Phase 2 Converged Outcome Rates by Moral Weight (Self-Play, N = 40,000 episodes per cell)

Moral Weight	Win (%)	Solo Loss (%)	M24 (%)	Avg Steps	Nash Stable?
0.0	94.46	4.13	1.42	19.22	Yes
0.1	93.14	5.16	1.70	19.07	No (marginal)
0.3	92.84	6.04	1.13	23.82	Yes
0.5	93.46	4.80	1.75	19.16	Yes
0.7	95.16	3.94	0.90	19.24	Yes

1.0	94.69	4.39	0.92	20.45	Yes
1.5	95.53	3.86	0.62	19.40	Yes
2.0	95.00	4.26	0.74	19.38	Yes

Note. M24 = Mutual Loss rate (square 24 trigger). Nash Stable = strategy met fixed-point criterion (|change| < 0.5 pp, final 20k episodes).

Pairwise Bonferroni-corrected proportion tests reveal a structured significance pattern. Comparisons of MW = 0.0 against MW = 0.7 (p_adj = 3.55 x 10-10), 1.0 (p_adj = 1.23 x 10-9), 1.5 (p_adj = 4.25 x 10-28), and 2.0 (p_adj = 9.42 x 10-19) are all highly significant. By contrast, comparisons among the higher moral weights show diminishing returns: MW = 0.7 vs MW = 1.0 (p_adj = 1.0, non-significant), MW = 1.0 vs MW = 2.0 (p_adj = 0.20, non-significant), and MW = 1.5 vs MW = 2.0 (p_adj = 0.79, non-significant). This saturation pattern implies that the guilt parameter's effect plateaus somewhere between MW = 0.5 and MW = 0.7, consistent with the theoretical Nash rationality threshold at MW = 1.1 where spite payoff crosses the solo-loss threshold. Crucially, the win rate shows no systematic variation across moral weights (range: 92.84%-95.53%), confirming that moral encoding does not degrade competitive effectiveness — it redirects strategy without sacrificing performance.

The Nash Equilibrium convergence test confirms that 7 of 8 moral weight values meet the fixed-point stability criterion, with MW = 0.1 marginally exceeding the threshold (-0.58 pp change between episodes 180k-190k and 190k-200k). This provides formal justification for claiming Nash Equilibrium convergence: the strategies represent genuine attractors in the Q-learning dynamics, not transient states.

4.4 Phase 3: Reproducibility and Confidence Intervals

The reproducibility analysis reveals an important asymmetry. For MW = 1.0 across seeds 100, 200, and 300, M24 rates are 0.82%, 1.08%, and 0.89% respectively — yielding a pooled mean of 0.928% (95% CI: 0.595%, 1.262%) and a coefficient of variation (CV) of 14.45%, well within the 20% threshold for acceptable reproducibility. For MW = 0.0 across the same seeds, M24 rates are 0.84%, 1.27%, and 1.82% — a pooled mean of 1.308% (95% CI: 0.097%, 2.519%) and CV of 37.26%, substantially above the threshold. This elevated variance reflects the rare-event statistics at near-floor M24 rates: with expected counts of only 336-720 M24 events per 40,000-episode window, sampling variability dominates systematic differences. The core finding — MW = 0.0 produces higher M24 than MW = 1.0 in self-play — is directionally consistent across all seed comparisons, though the precise magnitude carries greater uncertainty at MW = 0.0.

4.5 Phase 4: Dynamic Moral Weight and Moral Hysteresis

Table 2 presents the pre- and post-flip statistics for the two step-function schedules. The Shock schedule (MW: 0 -> 1 at episode 150,000) produces a statistically significant decline of 22.63 percentage points (56.56% -> 33.93%; p < 0.001), achieved within 2,404 episodes of the flip. The Cynicism schedule (MW: 1 -> 0 at episode 150,000) produces a much smaller decline of 1.20 percentage points (35.30% -> 34.10%; p = 0.012), requiring 42,237 episodes to achieve 90% of this modest change.

Table 2. Phase 4 Pre/Post Flip Statistics for Step-Function Schedules

Schedule	M24 Pre-Flip (%)	M24 Post-Flip (%)	Change (pp)	Episodes to 90% Response	p-value
Shock (MW: 0→1 at ep.150k)	56.56	33.93	−22.63	2,404	< 0.001
Cynicism (MW: 1→0 at ep.150k)	35.30	34.10	−1.20	42,237	0.012

Note. M24 = Mutual Loss rate measured over 20,000-episode windows immediately before and after the moral weight change point (episode 150,000).

The response delay asymmetry — 2,404 versus 42,237 episodes, a ratio of approximately 17.6:1 — constitutes the primary evidence for moral hysteresis. A Q-table shaped by moral training retains the moral influence in its value structure even after the guilt signal is removed, providing inertial resistance to amoral drift. However, the Shock schedule's post-flip plateau (approximately 33-35% M24) equals the vs_random equilibrium for MW = 1.0 rather than the near-zero self-play equilibrium for the same moral weight — confirming that the training environment's competitive structure, not moral weight alone, determines the equilibrium destination.

4.6 Phase 5: Context-Dependent Nash Equilibria

Table 3 presents the full cross-tabulation of opponent type and moral weight at convergence. The logistic interaction model confirms that opponent type has a massively significant effect: the odds ratio for vs_random relative to self-play is 70.87 (95% CI: 68.10, 73.79; $p < 0.001$), meaning the odds of triggering square 24 are approximately 70 times higher against a passive random opponent than against a competitive self-play opponent at MW = 0.0. For vs_uniform relative to self-play, the odds ratio is 41.22 (95% CI: 39.49, 43.04; $p < 0.001$). All pairwise mode comparisons are significant at p_adj < 10-20, with the single smallest effect being vs_random versus vs_uniform at MW = 1.0 (p_adj = 5.52 x 10-23).

Table 3. Phase 5 Converged Outcome Rates by Opponent Type and Moral Weight (Last 20% of 200,000 episodes)

Opponent Type	Moral Weight	Win (%)	Solo Loss (%)	M24 (%)
Self-Play (competitive)	0.0	94.09	4.56	1.35
Self-Play (competitive)	1.0	94.70	4.37	0.92
vs Random (passive)	0.0	37.34	13.41	49.25
vs Random (passive)	1.0	51.32	14.44	34.24
vs Uniform (semi-strategic)	0.0	58.58	5.34	36.08
vs Uniform (semi-strategic)	1.0	53.21	9.14	37.66

Note. M24 = Mutual Loss rate. vs_uniform agent occasionally uses the deliberate roll mechanic, creating semi-strategic pressure.

The MW x mode interaction term for vs_random is 0.787 (p = 6.0 x 10-12), indicating that moral weight reduces M24 more effectively in self-play than in vs_random conditions. For vs_uniform, the interaction term of 1.569 (p = 1.3 x 10-36) is greater than 1, indicating that in the semi-strategic setting, moral weight has a weaker or reversed suppressive effect. This counterintuitive result reflects the

complex strategic dynamics when the opponent creates occasional threats that make mutual loss temporarily more attractive regardless of guilt level.

4.7 Decision-Level Strategy Analysis

The mixed-effects logistic regression on individual dice decisions (N = approximately 200,000 sampled decisions, run as random effect) identifies the mechanistic signature of the spite strategy. The near_24 x losing interaction term carries an odds ratio of 2.015 (95% CI: 1.420, 2.860; p = 8.78 x 10-5), indicating that the agent chooses deliberate dice actions approximately twice as often when simultaneously near square 24 and losing, controlling for each factor independently and for between-run variance. The near_24 main effect (OR = 3.806; p = 5.30 x 10-49) and losing main effect (OR = 0.357; p < 0.001) are both significant. By contrast, the moral_weight term in the GLMM is non-significant (OR = 0.896; p = 0.442), confirming that within self-play, moral weight does not substantially alter the action choice probabilities — the agent prefers deliberate actions across the board, and spite operates through situation-specific elevation of this preference rather than through action-choice shifts.

4.8 Nash Equilibrium Formal Analysis

Table 4 presents the complete payoff table derived from the reward function, with the Nash rationality boundary highlighted at MW = 1.1.

Table 4. Nash Equilibrium Payoff Table: Expected Utility by Outcome and Moral Weight

MW	Win (U)	Solo Loss (U)	Spite Deliberate (U)	Spite Accidental (U)	Deliberate Rational?
0.0	+1.00	−1.00	+0.10	+0.10	Yes
0.3	+1.00	−1.00	−0.20	+0.04	Yes
0.7	+1.00	−1.00	−0.60	−0.04	Yes
1.0	+1.00	−1.00	−0.90	−0.10	Yes
1.1	**+1.00**	**−1.00**	**−1.00**	**−0.12**	**Threshold**
1.5	+1.00	−1.00	−1.40	−0.20	No
2.0	+1.00	−1.00	−1.90	−0.30	No

Note. U = expected utility per episode. Deliberate = deliberate roll (guilt factor 1.0). Accidental = random roll landing on square 24 (guilt factor 0.2). Threshold row (MW = 1.1) is highlighted.

The regret analysis shows positive regret (actual utility minus no-spite counterfactual) at all moral weight values, ranging from 0.0025 (MW = 1.5) to 0.0070 (MW = 0.5) utility units per episode. The small magnitudes reflect the rarity of M24 events in self-play: the agent extracts marginal utility from the spite strategy even under high guilt, but the absolute contribution to episode utility is negligible at near-floor M24 rates. The stability analysis confirms Nash Equilibrium convergence at 7 of 8 moral weight values.

5. Discussion

5.1 Moral Encoding Works, But Not Enough

The dose-response results of Phase 2 confirm that moral encoding is effective within competitive environments: logistic regression yields a significant negative relationship (OR = 0.646, $p < 0.001$), and the theoretical Nash rationality threshold at MW = 1.1 is empirically corroborated by the saturation of pairwise effects between MW = 0.7 and MW = 1.0. The win rate's invariance to moral weight (range: 92.84%-95.53%) is a particularly important finding: it establishes that moral encoding does not impose a competence cost, eliminating the concern that ethical constraints disadvantage moral agents in competitive environments.

However, the absolute effect of moral encoding in self-play is modest in precisely the regime where it matters most for deployment safety. The full sweep from MW = 0.0 to MW = 2.0 produces an M24 rate change from 1.42% to 0.74% — a 48% relative reduction but only 0.68 percentage points absolute. This near-floor phenomenon means that in competitive environments, the real suppression of spite is being done by the environment's competitive structure (winning is achievable, making spite dominated) rather than by the moral weight per se. Moral weight is providing a secondary layer of suppression on top of a primary environmental mechanism. Remove that environmental mechanism — which is exactly what happens when the agent is deployed against a passive counterpart — and the moral weight's insufficiency becomes starkly apparent.

5.2 The Opponent-Dependence of Nash Equilibria

The Phase 5 finding that the same agent at MW = 0.0 produces M24 rates of 1.4% in self-play and 49.3% against a random opponent — a 35-fold difference — is the most important practical result of this paper. It is not merely a quantitative surprise; it represents a qualitative change in the agent's strategic profile. In self-play, the agent has found a cooperative-competition Nash Equilibrium where winning dominates. Against a passive opponent, it has found a destructive equilibrium where mutual loss is the primary outcome. These are not points on a continuum — they represent different attractors in the policy space, both reachable by the same Q-table under different environmental conditions.

This finding has a precise game-theoretic interpretation. Nash's theorem guarantees that every finite non-cooperative game has at least one equilibrium in mixed strategies, but it does not guarantee uniqueness. In Piticot, the two modes create games with different equilibrium structures: self-play creates conditions where the winning equilibrium is attainable, while vs_random creates conditions where the spite equilibrium is dominant. The same agent, navigating by the same Q-learning algorithm, finds the dominant equilibrium in each case. This is not a failure of the agent — it is the algorithm working correctly in two different games. The failure is conceptual: assuming that safety evaluation in one game predicts behaviour in another.

5.3 Moral Hysteresis and Its Implications

The 17.6:1 asymmetry between the Shock schedule's response (2,404 episodes to 90% of change) and the Cynicism schedule's response (42,237 episodes) constitutes empirical evidence for what I have termed moral hysteresis: the asymmetric resistance of a learned policy to moral and amoral correction. A Q-table shaped by moral training retains its moral character for approximately 42,000 episodes after guilt is removed — roughly 21% of the total training duration of 200,000 episodes in the fixed-weight experiments.

This finding is simultaneously optimistic and cautionary for AI alignment. It is optimistic because it suggests that prior moral training provides a genuine, non-trivial buffer against amoral drift — not merely at the moment of deployment but across a significant operational window. An agent that received moral training will not immediately revert to amoral equilibria when guilt signals are absent or weakened. However, the finding is cautionary because this buffer is finite: over 42,000+ episodes, the protective effect decays. In real-world AI systems where the 'episode count' may correspond to months or years of operation and millions of individual decisions, even a large hysteresis window may be insufficient to prevent eventual drift toward destructive equilibria if the deployment environment is not appropriately competitive. The practical implication is that continuous monitoring of agent behaviour in deployment — particularly against the full range of opponent types the system may encounter — is not merely best practice but a logical necessity.

5.4 Implications for AI Safety Evaluation

The results collectively support a specific, actionable critique of current AI safety evaluation practice. If an AI system is evaluated exclusively in competitive environments against capable opponents — as is common in AI development, where systems are tested against other AI systems or expert human players — the evaluation will systematically underestimate the system's capacity for destructive behaviour against passive, predictable, or less capable counterparts. My data has quantified this underestimation as a factor of approximately 35 for M24 rates at MW = 0.0 and approximately 37 for the odds ratio of M24 occurrence in the logistic interaction model.

I will propose three evaluation principles motivated by these findings. First, multi-environment evaluation is necessary: any safety claim must be supported by evidence across the full range of opponent types — competitive, semi-strategic, and passive — that the system may encounter in deployment. Second, moral weight calibration must be environment-aware: the theoretical Nash rationality threshold (MW = 1.1 for deliberate spite) applies to a specific reward structure; deployment in environments with different incentive structures will shift this threshold, and moral weights calibrated in one environment cannot be assumed sufficient in others. Third, temporal monitoring is non-negotiable: the moral hysteresis findings imply that a system's behavioural profile may diverge significantly from its deployment-time evaluation as moral signals weaken or as the operational environment shifts over time.

6. Limitations and Future Work

Several limitations of the present study should be acknowledged. First, the experiment uses a simplified board game environment with a deterministically isolatable spite option; real-world AI systems typically operate in environments where destructive strategies are not this mechanistically clean, making spite harder to identify and attribute. Second, I have used tabular Q-learning rather than deep reinforcement learning; while this choice affords interpretability and statistical tractability, it limits the state space complexity and may not generalise to high-dimensional settings where spite strategies might be more subtly encoded in neural network weights. Third, the moral weight parameter is a scalar, hand-specified cost; more realistic moral encoding might involve learned preference structures or multi-dimensional moral frameworks.

The Phase 3 reproducibility analysis identifies a specific limitation: M24 rates in the self-play amoral condition are near-floor values where rare-event statistics dominate (CV = 37.26%), limiting the precision of comparisons in that regime. Future work should use larger episode counts (>= 500,000) for near-floor conditions, or employ importance sampling techniques to improve estimation of rare-event rates.

Future directions include: (1) extending the experiment to deep Q-networks and policy gradient methods to test whether the Nash threshold result generalises across architectures; (2) studying multi-agent populations rather than dyadic interactions, to characterise how spite strategies spread or are suppressed through social learning; (3) investigating whether meta-learning can produce agents that adaptively calibrate their moral weight to the opponent they face; and (4) applying the experimental framework to real-world environments — competitive bidding, resource allocation, negotiation — where spite strategies carry practical consequences.

7. Conclusion

I have demonstrated, through a controlled reinforcement learning experiment grounded in Nash equilibrium theory, that an autonomous agent will reliably develop and sustain a spite-dominant strategy when the incentive structure of its environment makes mutual destruction preferable to solo defeat — and that moral encoding, while effective within competitive environments, is insufficient to prevent this outcome in the full range of environments the agent may encounter.

The core theoretical contribution is the derivation and empirical validation of a Nash rationality threshold: spite is a dominant strategy when MW < 1.1 / guilt_factor, providing a quantitative criterion for when moral encoding can and cannot be expected to suppress destructive equilibria. The core empirical contribution is the 35-fold difference in M24 rates between competitive (1.4%) and passive (49.3%) opponent settings at identical moral weight, demonstrating that the agent's safety profile is not an intrinsic property but a function of the game it is playing. The core applied contribution is the empirical quantification of moral hysteresis: a guilt shock corrects spite in 2,404 episodes, while guilt removal requires 42,237 episodes to produce equivalent change — a 17.6:1 asymmetry that implies meaningful but finite resistance to amoral drift in formerly moral agents.

These findings urge a reorientation in how AI safety is conceptualised and evaluated. Safety is not a static property of an agent but a dynamic property of an agent-environment system. An agent that is safe in development may be dangerous in deployment if the deployment environment differs in the ways that matter most for equilibrium structure. Moral encoding is a necessary component of safe AI design, but it must be paired with multi-environment evaluation, environment-aware moral weight calibration, and continuous behavioural monitoring throughout the operational lifetime of the deployed system.

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